8.6

(a)

$$\begin{cases} H_0: \sigma_m^2 = \sigma_F^2 \\ H_a: \sigma_m^2 \neq \sigma_F^2 \end{cases}$$

$\alpha = 0.05$

test statistic : $F = \dfrac{\hat{\sigma}_m^2}{\hat{\sigma}_F^2} \overset{H_0}{\sim} F(577-4,\ 1000-577-4)$

$RR = \{ F \geq F_{0.975}(573, 419) = 1.9678 \text{ or } F \leq F_{0.025}(573, 419) \}$

$F^* = \dfrac{\frac{97161.9174}{573}}{12.024} = 1.173 \notin RR$

$\Rightarrow$ Do not reject H_0, we don't have enough evidence to say that $\sigma_m^2 \neq \sigma_F^2$

(b)

$$\begin{cases} H_0: \sigma^2_{married} = \sigma^2_{single} \\ H_a: \sigma^2_{married} > \sigma^2_{single} \end{cases}$$

$\alpha = 0.05$

test statistic : $F = \dfrac{\hat{\sigma}^2_{married}}{\hat{\sigma}^2_{single}} \overset{H_0}{\sim} F(1000-400-5,\ 400-5)$

$R.R. = \{ F \geq F_{0.95}(595, 395) = 1.6470 \}$.

$F^* = \dfrac{\frac{100703.0471}{595}}{\frac{56231.0382}{395}} = 11.898 \in RR$

$\Rightarrow$ reject H_0, we have enough evidence to say that $\sigma^2_{married} > \sigma^2_{single}$.

(c) NR^2 test

Without intercept, $df = 4$

test statistic : $\chi^2 = N \times R^2 \sim \chi^2_4$

$R.R. = \{ \chi^2 \geq \chi^2_{0.05,4} = 9.488 \}$.

$\chi^2 = 59.03 \in RR$

$\Rightarrow$ reject H_0, we can't say the model is homoskedasticity.

證據表明模型存在異質變異數,但此檢定不能直接證明(b)所討論的「已婚&未婚者之間」有異質變異數。

原因為(c)的 Auxiliary regression 沒有包含 Married,所以只能說誤差變異數和這些變數有關,不能直接說異質變異數來自已婚&未婚的差異

(d) White test

$df = 4 + 2 + \binom{4}{2} = 12$

White test statistic : $\chi^2 = 78.82$

$RR = \{ \chi^2 \geq \chi^2_{0.95,12} = 21.03 \}$

$\chi^2 \in RR$

$\Rightarrow$ reject H_0, we can't say the model is homoskedasticity.

(e)

變窄 $\Rightarrow$ {EXPER, METRO, FEMALE} (usual > robust)

變寬 $\Rightarrow$ {Intercept, EDUC} (usual < robust)

異質變異數 robust standard error 並不一定都比一般 standard error 大
Robust SE 會根據殘差變異數和解釋變數的關係調整，所以即使檢定顯示有 heteroskedasticity，也不代表所有 robust SE 都一定要變大.

例

(b) 是 $\begin{cases} H_0: \sigma^2_{married} = \sigma^2_{single} \\ H_a: \sigma^2_{married} > \sigma^2_{single} \end{cases}$，也就是在看「兩者之變異是否不同」，屬於 heteroskedasticity 的問題

此小題將 MARRIED 加進 wage equation，然後看 t-value. $H_0: \beta_{Married} = 0$，這是在檢定「已婚狀態是否影響平均工資」，這是 conditional mean 的問題

∵ 即使 $t \approx 1$ 表 MARRIED 對平均工資不顯著，也仍可能存在 $\sigma^2_{married} > \sigma^2_{single}$

∴ 不衝突

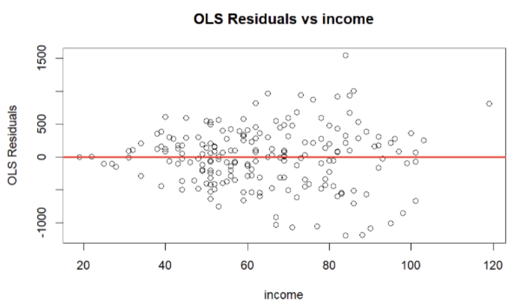
8.16

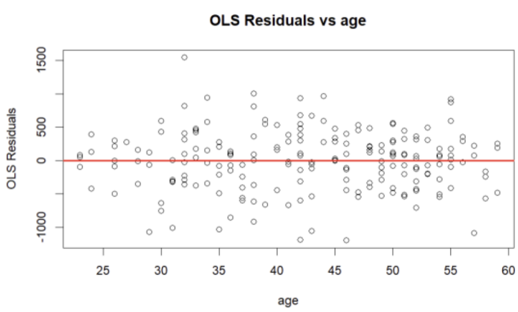
(a)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
     Min       1Q   Median       3Q      Max 
-1198.14  -295.31    17.98   287.54  1549.41 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -391.548    169.775  -2.306   0.0221 *  
income        14.201      1.800   7.889 2.10e-13 ***
age           15.741      3.757   4.189 4.23e-05 ***
kids         -81.826     27.130  -3.016   0.0029 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,	Adjusted R-squared:  0.3305 
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16

                 2.5 %    97.5 %
(Intercept) -726.36871 -56.72731
income        10.65097  17.75169
age            8.33086  23.15099
kids        -135.32981 -28.32302
```

```
          2.5 %    97.5 %
kids -135.3298 -28.32302
```

(b)





可以觀察到 income 的圖隨 X 上升，誤差越來越大，表示可能存在 heteroskedasticity

(c)

$$H_0: \sigma^2_{high\ income} = \sigma^2_{lowin\ come}$$

$$H_a: \sigma^2_{high\ income} > \sigma^2_{low\ income}$$

```
Call:
lm(formula = miles ~ income + age + kids, data = low_income)

Residuals:
    Min      1Q  Median      3Q     Max 
-684.07 -245.39    8.69  202.87  631.43 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -392.511    214.166  -1.833  0.07030 .  
income        10.960      3.770   2.907  0.00464 ** 
age           18.869      3.783   4.988 3.14e-06 ***
kids         -70.371     29.138  -2.415  0.01785 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 319 on 86 degrees of freedom
Multiple R-squared:  0.309,	Adjusted R-squared:  0.2849 
F-statistic: 12.82 on 3 and 86 DF,  p-value: 5.31e-07


Call:
lm(formula = miles ~ income + age + kids, data = high_income)

Residuals:
     Min       1Q   Median       3Q      Max 
-1215.44  -426.21    73.56   304.71  1602.70 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept) -476.803    548.833  -0.869   0.3874   
income        15.556      5.450   2.855   0.0054 **
age           16.388      7.385   2.219   0.0291 * 
kids        -116.017     49.861  -2.327   0.0223 * 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 562 on 86 degrees of freedom
Multiple R-squared:  0.1514,	Adjusted R-squared:  0.1218 
F-statistic: 5.116 on 3 and 86 DF,  p-value: 0.002642
```

F_stat = 3.104061

F_crit = 1.428617

p_value = 1.64001e-07

Reject H0: heteroskedasticity is present.

(d)

```
t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
income        14.2013     1.9389  7.3246 6.083e-12 ***
age           15.7409     3.9657  3.9692 0.0001011 ***
kids         -81.8264    29.1544 -2.8067 0.0055112 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

coefficient <dbl>	robust_se <dbl>	lower_95 <dbl>	upper_95 <dbl>
-81.82642	29.15438	-139.323	-24.32986

(e)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/income^2)

Weighted Residuals:
     Min       1Q   Median       3Q      Max
-15.1907  -4.9555   0.2488   4.3832  18.5462

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -424.996    121.444  -3.500 0.000577 ***
income        13.947      1.481   9.420  < 2e-16 ***
age           16.717      3.025   5.527 1.03e-07 ***
kids         -76.806     21.848  -3.515 0.000545 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.765 on 196 degrees of freedom
Multiple R-squared:  0.4573,    Adjusted R-squared:  0.449
F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

KIDS 的 conventional GLS CI

```
    2.5 %    97.5 %
-119.8945 -33.71808
```

```
t test of coefficients:

             Estimate Std. Error t value  Pr(>|t|)
(Intercept) -424.9962    95.8035 -4.4361 1.526e-05 ***
income        13.9473     1.3470 10.3545 < 2.2e-16 ***
age           16.7175     2.7974  5.9761 1.061e-08 ***
kids         -76.8063    22.6186 -3.3957 0.0008286 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

KIDS 的 robust GLS CI

```
-121.41339  -32.19919
```

Method <chr>	Coefficient <dbl>	SE <dbl>	Lower_95 <dbl>	Upper_95 <dbl>
OLS conventional	-81.82642	27.12960	-135.3298	-28.32302
OLS robust	-81.82642	29.15438	-139.3230	-24.32986
GLS conventional	-76.80629	21.84844	-119.8945	-33.71808
GLS robust	-76.80629	22.61861	-121.4134	-32.19919